

Zohreh Mostaani

Machine Learning and Data Scientist

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Summary

Machine Learning Engineer/Researcher with 7+ years building ML pipelines, covering data acquisition/curation, signal preprocessing, modeling, experimentation, and rigorous evaluation, across radar, speech, and physiological time series. Built and validated automated evaluation frameworks for generative speech (TTS), benchmarked against human listening studies. Strong in Python, PyTorch/TensorFlow, and Docker; experienced collaborating with cross-functional stakeholders. Delivered models for detection, prediction, and biometric analytics; created datasets; and mentored junior contributors. 15+ publications (~600 citations, h-index 9).

Skills

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- Languages: Python (NumPy, Pandas, SciPy), Shell; MATLAB
- Deep learning: PyTorch, TensorFlow, scikit-learn; CNN/RNN; transformers/attention; time-series modeling; feature engineering; evaluation/ablation; TTS/speech synthesis evaluation; speech quality metrics (MOS)
- Systems & MLOps: Docker, Git; environment management (conda/mamba, Poetry, uv); REST APIs (FastAPI/Flask); AWS, Azure
- Data: dataset design/collection, labeling protocols, data loaders/pipelines, signal processing, visualization (Matplotlib/Seaborn/Plotly); human evaluation / A/B testing
- Collaboration: experiment tracking/reporting, mentoring, cross-functional work
- Languages: English (fluent), French (intermediate), Persian (native)

Experience

AGIGO AG	Zürich, Switzerland
Research Engineer	Jan 2026 -- Present

Generative speech evaluation for an agentic, LLM-driven conversational AI platform.

- Built an automated evaluation framework to benchmark 9 text-to-speech (TTS) voices across 15 domains, producing a single quality score from intelligibility, audio quality, speaker similarity, and synthesis success rate, reducing reliance on manual listening tests. Validated against a human ranking study: automatic and human rankings agreed on the top-3 and bottom-2 voices, with minor reordering among the middle ranks.
- Designed a diagnostic report that localized TTS errors via an ASR round-trip, classifying mismatches by severity so developers could trace each one to the TTS, ASR, or normalization stage.
- Deployed the framework as a Dockerized microservice exposing HTTP endpoints, integrating it into the team's model-evaluation workflow.

Sleepiz AG	Zürich, Switzerland
Machine Learning Engineer (BNF Project)	Jul 2025 -- Oct 2025

Radar signal processing and ML pipeline

- Designing and implementing algorithms for noncontact human detection and vitalsign monitoring using radar signals.

Idiap Research Institute	Martigny, Switzerland
Research Assistant	Feb 2020 -- Dec 2024
Integrated physiological signals in speech processing technologies using machine learning and deep learning.	
•Pioneered developing deep learning models to estimate breathing signals from raw speech waveforms using TensorFlow. Achieved a 25% reduction in input lag compared to spectral-based models. •Developed models for COVID-19 detection, presentation attack detection, emotion recognition, hypoglycemia detection in diabetic patients, and heart rate estimation in PyTorch and Scikit-learn using data-driven and hand-crafted features extracted from speech signals. •Compiled two databases with simultaneous recordings of speech and physiological signals from more than 30 participants. •Guided two Master's students through internships and thesis projects related to physiological signal processing. Supported Master's level courses <i>Machine Learning</i> and <i>Introduction to Speech Processing</i> as a teaching assistant, providing help with assignments, projects, coding challenges, and grading.	

Idiap Research Institute	Martigny, Switzerland
Research and Development Engineer	Apr 2019 -- Jan 2020
•Documented, tested, and added new functionalities to BEAT framework; a framework enabling the definition, execution, and comparison of data-driven workflows: Python, Django, Sphinx. •Devised a protocol for collecting face biometric data in automotive environments.	

Idiap Research Institute	Martigny, Switzerland
Internships	Jun 2016 -- Mar 2019
•Collected two extensive face biometric datasets from over 100 participants, including protocol design for data acquisition and generation of face presentation attacks. •Incorporated an Automatic Speech Recognition <i>ASR</i> system into a web-based framework using Docker and RESTful API. •Implemented a Convolutional Neural Network for <i>Gaze Estimation</i> in Theano using data collected at Idiap.	

Özyeğin University	Istanbul, Turkey
Research Assistant	Feb 2012 -- Jan 2015
•Enhanced system efficiency in free-space optical communication by achieving a 50% improvement in signal-to-noise ratio (SNR) through the use of multiple relays. •Managed assignments, labs, and exams for Bachelor's and Master's level courses, including Physics, Mathematics, Linear Algebra, and Wireless Communications.	

Education

EPFL University	PhD
Electrical and Electronics Engineering	Mar 2025
Özyeğin University	M.Sc.
Electrical and Electronics Engineering	Jan 2015
University of Tehran	B.Sc.
Electrical Engineering	Sep 2011